1) **Write a ‘C’ program to interchange two numbers without using third variable.**

Ans-

```
#include <stdio.h>
int main()
{
    int a, b;

    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);

    a = a + b;
    b = a - b;
    a = a - b;

    printf("After interchange: a = %d, b = %d", a, b);

    return 0;
}
```

O/P - Enter two numbers: 10 20
After interchange: a = 20, b = 10

2) **Write a function to display first 10 prime numbers.**

Ans -

```
#include <stdio.h>
void prime()
{
    int n, i, count = 0;

    for (n = 2; count < 10; n++)
    {
        for (i = 2; i < n; i++)
        {
            if (n % i == 0)
                break;
        }

        if (i == n)
        {
            printf("%d ", n);
```

```
            count++;
        }
    }
}

int main()
{
    prime();
    return 0;
}
```

O/P - 2 3 5 7 11 13 17 19 23 29

3) Write a program to display the following pattern.

1 2 3
1 2
1

Ans -

```
#include <stdio.h>

int main()
{
    int i, j;

    for (i = 3; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }

    return 0;
}
```

O/P –

1 2 3
1 2
1

4) Write a ‘C’ program to perform the following operations on an integer till user chooses EXIT.
1. Check number is even or odd.
2. Display last digit of given number
3. EXIT

Ans -

```
#include <stdio.h>
int main()
{
    int n, ch;

    do
    {
        printf("\n1.Even/Odd\n2.Last Digit\n3.EXIT\n");
        printf("Enter choice: ");
        scanf("%d", &ch);

        switch (ch)
        {
            case 1:
                printf("Enter number: ");
                scanf("%d", &n);
                if (n % 2 == 0)
                    printf("Even");
                else
                    printf("Odd");
                break;

            case 2:
                printf("Enter number: ");
                scanf("%d", &n);
                printf("Last digit = %d", n % 10);
                break;

            case 3:
                break;

            default:
                printf("Invalid choice");
        }
    } while (ch != 3);
```

```
        return 0;
    }
```

O/P-1.Even/Odd
2.Last Digit
3.EXIT
Enter choice: 1
Enter number: 8
Even

1.Even/Odd
2.Last Digit
3.EXIT
Enter choice: 2
Enter number: 456
Last digit = 6

1.Even/Odd
2.Last Digit
3.EXIT
Enter choice: 3

5) Write a 'C' program which accepts a character and integer n as parameter and displays the next n characters.

Ans -

```
#include <stdio.h>
void display(char ch, int n)
{
    int i;

    for (i = 1; i <= n; i++)
    {
        ch++;
        printf("%c ", ch);
    }
}

int main()
{
    char ch;
    int n;
```

```
    printf("Enter a character: ");
    scanf(" %c", &ch);

    printf("Enter n: ");
    scanf("%d", &n);

    display(ch, n);

    return 0;
}
```

O/P- Enter a character: A
Enter n: 5
B C D E F

6) Write a 'C' program, which accepts two integers and an operator as a character (+ - * /), performs the corresponding operation and displays the result. (hint : use switch_case)

Ans -
```
#include <stdio.h>
    int main()
{
    int a, b;
    char op;

    printf("Enter two integers: ");
    scanf("%d%d", &a, &b);

    printf("Enter operator (+ - * /): ");
    scanf(" %c", &op);

    switch (op)
    {
        case '+': printf("Result = %d", a + b); break;
        case '-': printf("Result = %d", a - b); break;
        case '*': printf("Result = %d", a * b); break;
        case '/':
            if (b != 0)
                printf("Result = %d", a / b);
            else
                printf("Division by zero not allowed");
            break;
        default: printf("Invalid operator");
```

```
    }

    return 0;
}
```

O/P - Enter two integers: 10 5
Enter operator (+ - * /): +
Result = 15